

Unit 12

NITROGEN AND SULFUR

20 items · 47 marks

Syllabus sections in this unit

12.1 Nitrogen and sulfur

Unit 12: Nitrogen and sulfur

1. M/J 26/P21/Q2(a)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

(a) Write an overall equation for stage 3.

[1]

2. M/J 26/P21/Q2(c)(i)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (c) (introduction)

(c) CaCO_3 and O_2 are added to the waste gases from the Contact process before they escape. This prevents the release of SO_2 into the atmosphere.



(i) State **one** environmental consequence of the release of $\text{SO}_2(\text{g})$ into the atmosphere.

[1]

3. M/J 26/P24/Q2(a)(ii)

(ii) Explain the lack of reactivity of nitrogen.

[2]

4. MAR 26/P22/Q2(c)

Context

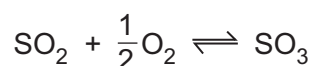
2 Nitrogen monoxide, NO, is an atmospheric pollutant.**(c) (i)** State a natural process that creates atmospheric NO.

[1]

(ii) Photochemical smog forms when NO reacts with unburned hydrocarbons.

Identify the major component of photochemical smog.

[1]

(iii) NO has a catalytic role in the oxidation of atmospheric SO₂.

Show, using relevant equations, the catalytic role that NO has in this oxidation.

[2]

5. M/J 25/P23/Q3(a)(ii)

Context: part (a) (introduction)

3 (a) Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.**(ii)** Describe how photochemical smog is produced during the combustion of petrol in an internal combustion engine.

[2]

6. O/N 24/P22/Q3(a)(i)-(iii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (a) (introduction)

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.**(i)** Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural

man-made

[2]

(ii) Write an equation to describe the role of NO₂ in the direct formation of acid rain.

[1]

(iii) Peroxyacetyl nitrate, PAN, is a component of photochemical smog.Describe how PAN forms from NO₂.

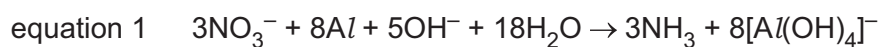
[1]

7. O/N 24/P22/Q3(b)(iii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al. Equation 1 shows the reaction.**(iii)** NH_3 is a basic gas.Describe how NH_3 is able to act as a base.

[1]

8. O/N 24/P22/Q3(f)(ii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (f) (introduction)

(f) At very high temperatures, phosphorus can form P_2 molecules. P_2 contains a triple bond, $\text{P}\equiv\text{P}$.**(ii)** The bond energy of $\text{P}\equiv\text{P}$ is 485kJ mol^{-1} . The bond energy of $\text{N}\equiv\text{N}$ is 944kJ mol^{-1} . Compare the reactivity of P_2 and N_2 . Explain your answer.

[1]

9. M/J 24/P23/Q1(a)(i)**1 (a) (i)** Explain the lack of reactivity of nitrogen gas, $\text{N}_2(\text{g})$.

[2]

10. M/J 24/P23/Q2(d)

Context

2 $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

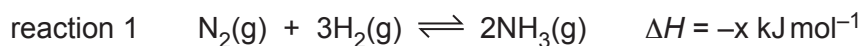


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

(d) When oxides of nitrogen escape into the atmosphere they may be involved in:

- formation of acid rain from sulfur dioxide
- formation of photochemical smog.

(i) Identify the role of NO and NO_2 in the formation of H_2SO_4 from SO_2 in the atmosphere to produce acid rain.

Use relevant equations to support your answer.

[3]

(ii) Outline how NO and NO_2 may contribute to the formation of photochemical smog.

[2]

11. M/J 24/P23/Q4(b)(ii)

Context: part (b) (introduction)

(b) An exothermic reaction occurs when $\text{NH}_4^+(\text{aq})$ is added to $\text{OH}^-(\text{aq})$.

(ii) Construct an ionic equation for the reaction of NH_4^+ and OH^- .

[1]

12. MAR 24/P22/Q3(a)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1

2

[2]

13. MAR 24/P22/Q3(b)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(b) N_2 can react with oxygen in an internal combustion engine to form a mixture of NO and NO_2 . Fig. 3.1 shows a reaction scheme involving N_2 .

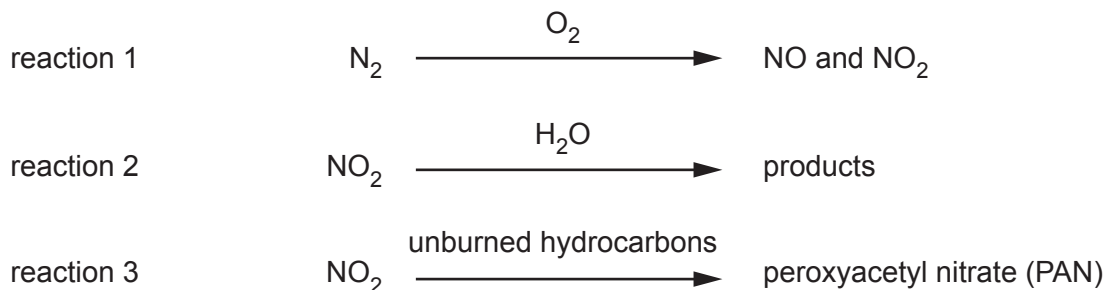


Fig. 3.1

(i) Write an equation to show the formation of a mixture of NO and NO_2 in reaction 1.

[1]

(ii) Give the formulae of the products of reaction 2.

[1]

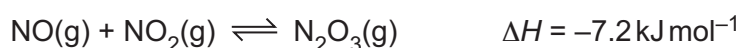
(iii) State **one** environmental consequence of reaction 3.

[1]

14. O/N 23/P22/Q2(d)

Context

2 NO and NO_2 react at 25 °C to give N_2O_3 as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

(d) Identify one natural process and one man-made process that cause the formation of atmospheric NO and NO_2 .

natural process

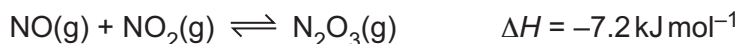
man-made process

[2]

15. O/N 23/P22/Q2(e)(ii)-(iii)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

Context: part (e) (introduction)

- (e)** NO₂ is a brown gas that can be used to form nitric acid.

- (ii)** NO₂ has a catalytic role in the oxidation of atmospheric sulfur dioxide.

Write equations to show the catalytic role of NO₂ in this oxidation.

[2]

- (iii)** State **one** environmental consequence of the oxidation of atmospheric sulfur dioxide.

[1]

16. M/J 23/P21/Q5(b)

Context: part (a) (introduction)

- 5** **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

- (b)** When petrol is burned in an internal combustion engine, oxides of nitrogen are released into the atmosphere. Oxides of nitrogen are responsible for the formation of acid rain.

- (i)** Suggest the conditions required for the production of oxides of nitrogen during combustion of **M** in an internal combustion engine. Use an appropriate equation in your answer.

[2]

- (ii)** Describe how acid rain is formed in the atmosphere in the presence of oxides of nitrogen and SO₂. Identify the role of the oxides of nitrogen in this process. Include **all** relevant equations.

[3]

- (iii)** State **one** other type of air pollution that is caused by the production of oxides of nitrogen in an internal combustion engine.

[1]

17. O/N 22/P22/Q1(c)(ii)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

Context: part (c) (introduction)

(c) NH_4^+ is a Brønsted–Lowry acid.

(ii) When $\text{NH}_4^+(\text{aq})$ is heated with $\text{NaOH}(\text{aq})$, a pungent gas is produced.

Write an ionic equation for this reaction.

[1]

18. M/J 22/P21/Q2(b)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(b) Nitrogen oxides, NO_2 and NO , are produced in internal combustion engines. Release of these gases into the atmosphere leads to the formation of photochemical smog.

(i) Outline how nitrogen oxides are involved in the formation of photochemical smog.

[2]

(ii) Construct an equation to demonstrate how a catalytic converter reduces the amount of nitrogen oxide gases released into the atmosphere.

[1]

19. M/J 22/P21/Q2(c)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(c) $\text{N}_2(\text{g})$ is very unreactive. It is difficult to make ammonia, $\text{NH}_3(\text{g})$, directly from its elements but it can be made from $\text{NH}_4\text{Cl}(\text{s})$.

Identify a reagent and the conditions required to make $\text{NH}_3(\text{g})$ from $\text{NH}_4\text{Cl}(\text{s})$.

[1]

20. MAR 22/P22/Q2(d)(i)-(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (d) (introduction)

(d) SO_2 and SO_3 are found in the atmosphere.

The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by NO_2 .

The first step of the catalytic oxidation is shown in equation 1.



(i) Construct an equation to show how NO_2 is regenerated in the catalytic oxidation of SO_2 .

[1]

(ii) NO_2 can also react with unburned hydrocarbons to form photochemical smog.

State the product of this reaction that contributes to photochemical smog.

[1]

Answers Section

1. M/J 26/P21/Q2(a)

2(a)	$\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$	1	BOD $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ BOD $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2 \text{H}_2\text{SO}_4$
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2. M/J 26/P21/Q2(c)(i)

2(c)(i)	acid rain	1	A breathing / respiratory problems / asthma A harmful to fish / plants / animals A damages buildings / statues / trees BOD (photochemical) smog I ozone I greenhouse effect / global warming / climate change I harmful to health
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3. M/J 26/P24/Q2(a)(ii)

2(a)(ii)	M1 (N_2 molecules / N_2) have a strong triple (covalent) bond OR $\text{N} \equiv \text{N}$ (bond) is strong M2 (N_2 molecules / N_2) are non-polar	2	M1 A strong = a lot of energy to break / high activation energy / high bond energy BOD hard/difficult to break I three strong (covalent) bonds (alone) A three strong bonds between N atoms R three strong bonds AND triple bond ($\therefore = 9$) M2 A same electronegativity / no difference in electronegativity
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4. MAR 26/P22/Q2(c)

2(c)(i)	lightning	1
2(c)(ii)	PAN / peroxyac(et)yl nitrate	1
2(c)(iii)	M1 first equation $\text{NO} + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2$ M2 second equation $\text{SO}_2 + \text{NO}_2 \rightarrow \text{SO}_3 + \text{NO}$	2

5. M/J 25/P23/Q3(a)(ii)

3(a)(ii)	- unburned hydrocarbons - react with NO / NO_2 (to) produce PAN (– a component of photochemical smog) Any 2 points = 1 mark, all 3 points = 2 marks	2
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6. O/N 24/P22/Q3(a)(i)-(iii)

3(a)(i)	(natural =) lightning	1
	(man-made =) internal combustion engines	1
3(a)(ii)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ OR $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$	1
3(a)(iii)	It / NO_2 reacts with (unburned) hydrocarbons / VOCs ALLOW reaction of unburned hydrocarbons / VOCs in presence of NO_2	1

7. O/N 24/P22/Q3(b)(iii)

3(b)(iii)	(NH_3 is an) H^+ acceptor	1
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8. O/N 24/P22/Q3(f)(ii)

3(f)(ii)	$P \equiv P$ is much weaker so P_2 is more reactive (than N_2)	1
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9. M/J 24/P23/Q1(a)(i)

1(a)(i)	M1 (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond M2 (N_2 molecules /molecule(s) of N_2) are non-polar	2
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10. M/J 24/P23/Q2(d)

2(d)(i)	M1 catalyst M2 2 equations to show the catalytic activity of NO_2 in formation of SO_3 from SO_2 $SO_2 + NO_2 \rightarrow NO + SO_3$ $2NO + O_2 \rightarrow 2NO_2$ M3 describe production of H_2SO_4 from SO_3 and water (can be shown in an equation). SO_3 then reacts with (rain)water / moisture / water vapour to make sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	3
2(d)(ii)	M1 unburned hydrocarbons combine with NO and / NO_2 M2 to form PAN / peroxyac(et)yl nitrate	2

11. M/J 24/P23/Q4(b)(ii)

4(b)(ii)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$	1
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12. MAR 24/P22/Q3(a)

3(a)	strong triple bond / high triple bond enthalpy	1
	non-polar (molecule)	1

13. MAR 24/P22/Q3(b)

3(b)(i)	e.g. $N_2 + 1\frac{1}{2}O_2 \rightarrow NO + NO_2$	1
3(b)(ii)	HNO_2 and HNO_3	1
3(b)(iii)	photochemical smog	1

14. O/N 23/P22/Q2(d)

2(d)	natural process: lightning	1
	man-made process: internal combustion engines	1

15. O/N 23/P22/Q2(e)(ii)-(iii)

2(e)(ii)	$NO_2 + SO_2 \rightarrow SO_3 + NO$	1
	$NO + \frac{1}{2}O_2 \rightarrow NO_2$	1
2(e)(iii)	(formation of) acid rain	1

16. M/J 23/P21/Q5(b)

5(b)(i)	M1 • high temperature • high pressure • N_2 Any two correct for M1 M2 $N_2 + O_2 \rightarrow 2NO$ OR $N_2 + 2O_2 \rightarrow 2NO_2$ OR $N_2 + xO_2 \rightarrow 2NO_x$	2
5(b)(ii)	M1 NO_2 is the catalyst (for $SO_2 \rightarrow SO_3$) OR catalytic oxidation (of SO_2) by NO_2 M2 $NO_2 + SO_2 \rightarrow NO + SO_3$ AND $2NO + O_2 \rightarrow 2NO_2$ M3 SO_3 reacts with rain OR moisture/water (in atmosphere) to form sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	3
5(b)(iii)	photochemical smog / PAN	1

17. O/N 22/P22/Q1(c)(ii)

1(c)(ii)	<i>ionic equation</i> $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$	1
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18. M/J 22/P21/Q2(b)

2(b)(i)	M1 react with (unburnt) hydrocarbons M2 (form) PAN / peroxyac(et)yl nitrate	2
2(b)(ii)	$2\text{NO} + 2\text{CO} \rightarrow 2\text{CO}_2 + \text{N}_2$ OR $\text{NO}_2 + 2\text{CO} \rightarrow \frac{1}{2}\text{N}_2 + 2\text{CO}_2$	1

19. M/J 22/P21/Q2(c)

2(c)	any Group 1 hydroxide or $\text{Ca}(\text{OH})_2$ / $\text{Sr}(\text{OH})_2$ / $\text{Ba}(\text{OH})_2$	1
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20. MAR 22/P22/Q2(d)(i)-(ii)

2(d)(i)	$\text{NO} + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2$	1
2(d)(ii)	peroxyac(et)ynitrate / PAN	1